

Two layouts, with the shared-room ventilation included

Full-house comparison 1:150 at A3 | South / garden up | Dashed lines are candidate overhead spine routes, not every branch

A / ONE CENTRAL UNIT



A / LARGER LINEN GAIN, VISIBLE SHARED CROSSING

Rear unit connects to both wings; amber crosses 13.05 m between halls.
Test a continuous 0.65 m projection x 0.35 m deep edge enclosure.
The band also runs along the garden-room / shared-room opening.
Conditional linen run: 2.05 m, after the old plant is fully relocated.

B / TWO LOCAL UNITS



B / NO THROUGH-RUNNING FAMILY TRUNKS

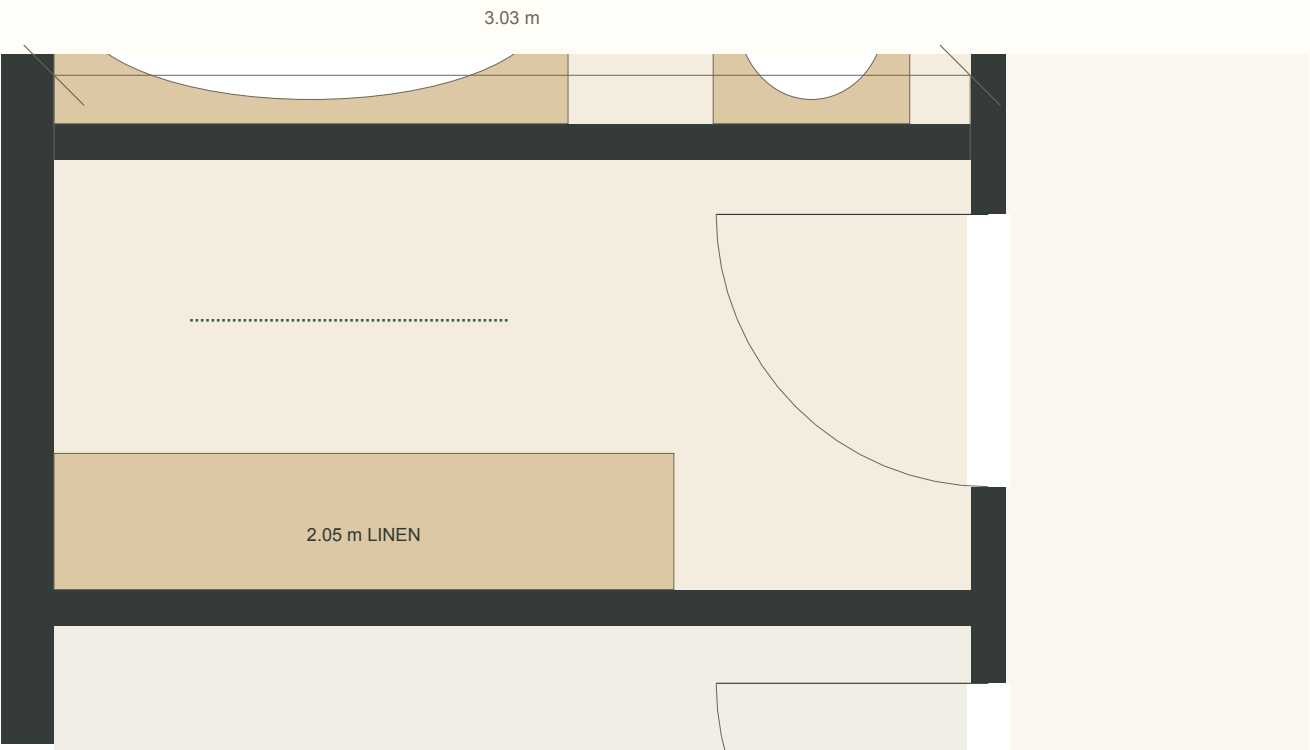
Family unit also supplies the living end from the family-wing boundary.
Service unit supplies dining / garden; kitchen extract uses the pantry route.
High-wall outlet reach must be proven; this is not a throw calculation.
Conditional linen run: 1.25 m, plus a 0.80 m-deep service bay.

Roof circles: intake / exhaust position allowances. All wet-room and bedroom branches, pressure losses, controls and terminal details still need design.

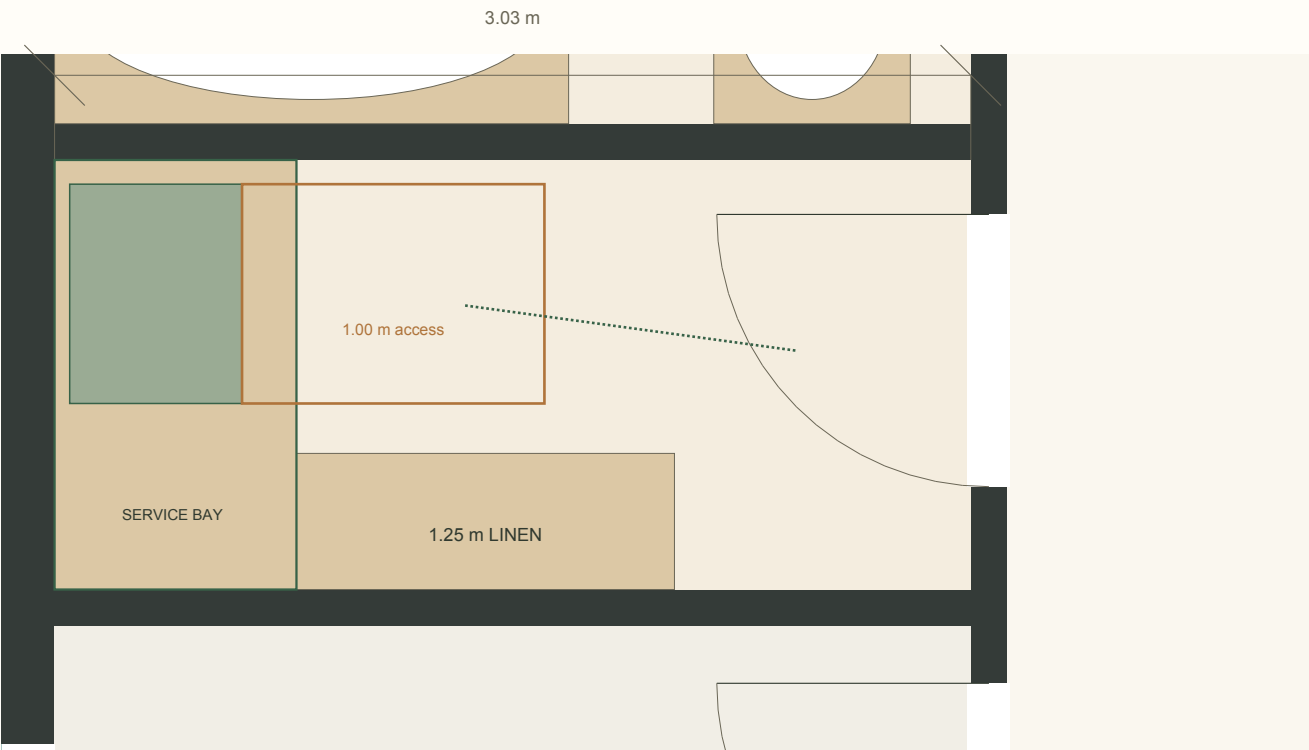
The family unit fits best facing along the cupboard

ST clear dimensions 3.03 x 1.42 m | Plans 1:25 at A3 | Both alternatives assume the original heating / water plant is relocated

A / LINEN ONLY



B / LINEN + MVHR



A / 2.05 m SHELF RUN

0.45 m deep; five illustrative shelf levels.
4.61 m2 total nominal shelf surface.
Existing proposal: 0.90 m run / 2.03 m2 shelf surface.
The shelf stops clear of the existing D04 door sweep.

B / 1.25 m SHELF RUN

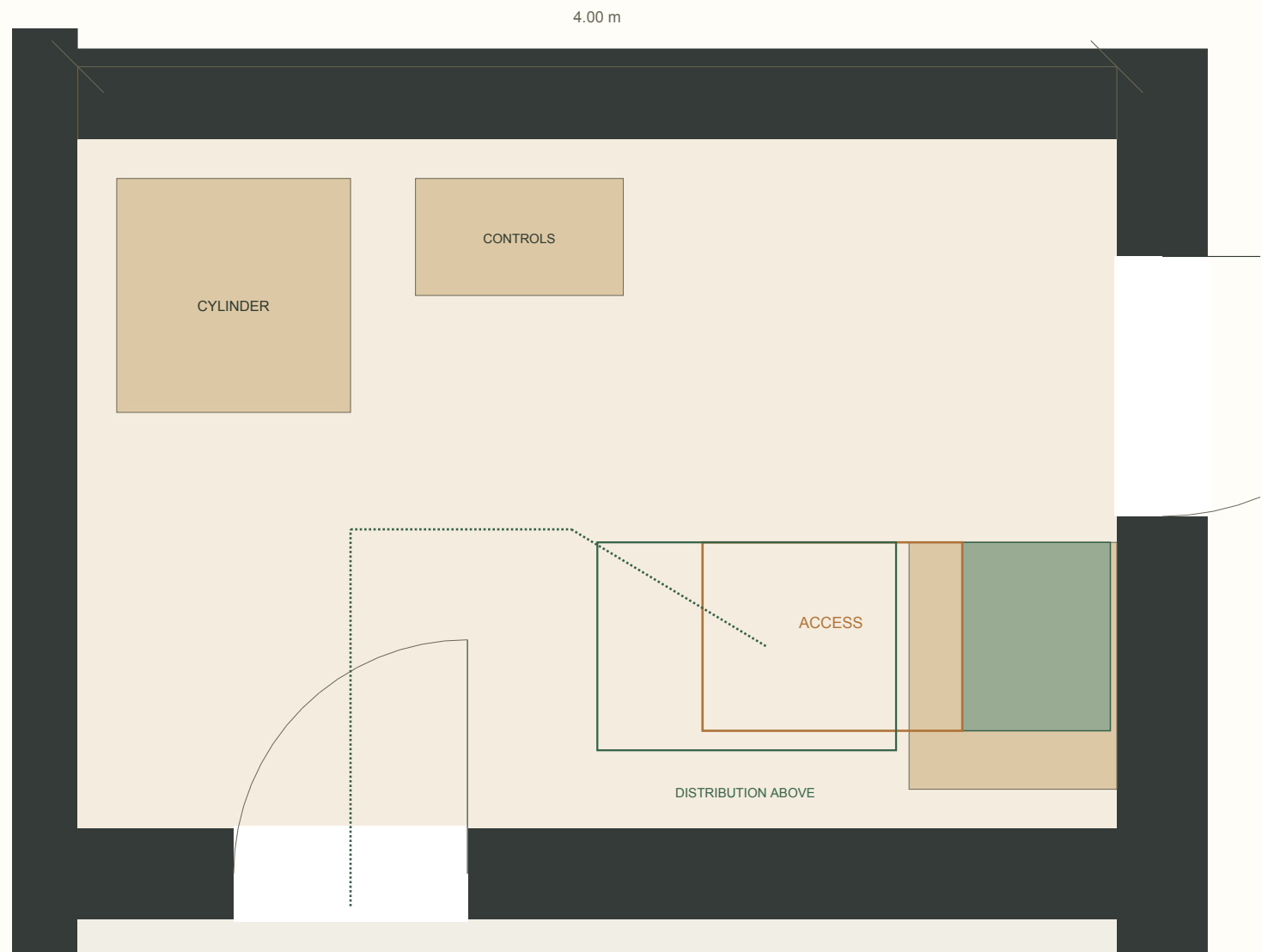
Same depth / levels: 2.81 m2 nominal shelf surface.
1.80 m2 less shelf surface than option A.
Still 0.35 m more shelf run than the existing proposal.
Unit faces the long room axis; front service space stays clear.

BODY FIT IS NOT A COMPLETE CUPBOARD SPECIFICATION

Q-series spatial example: 0.725 m wide x 0.570 m deep x about 0.850 m high; reserve 1.00 m in front [1, 2]. No capacity is selected.
Across the narrow room: 0.570 + 1.00 = 1.570 m, already more than 1.42 m before wall offsets or joinery. Rotating avoids that conflict.
The 0.80 x 1.42 m bay reserves connections above. Acoustic lining, access panels and any UFH manifold could reduce the shelf gain further.
Use a suitable support / isolation detail at the external wall; verify noise into adjacent bedrooms. Door handling and frames remain to check.

Retain a full maintenance approach in the rear plant room

Plant plan 1:25 at A3 | Same spatial unit envelope for either strategy; actual unit duty and model remain to select

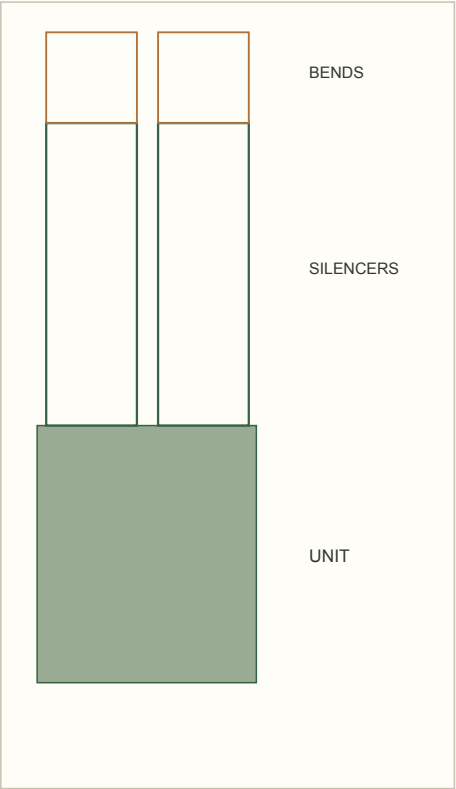


ACCESS AND CONNECTIONS

The 1.00 m front reservation clears fixed equipment and the open D21 leaf.
A 1.15 x 0.80 m ceiling distribution zone sits over clear service floor.
Four unit connections, outdoor duct insulation, drainage and supports must be coordinated in 3D before this becomes an installation layout.
The front access area cannot become storage; rear exterior door stays.

UNIT-WALL SECTION / 1:25

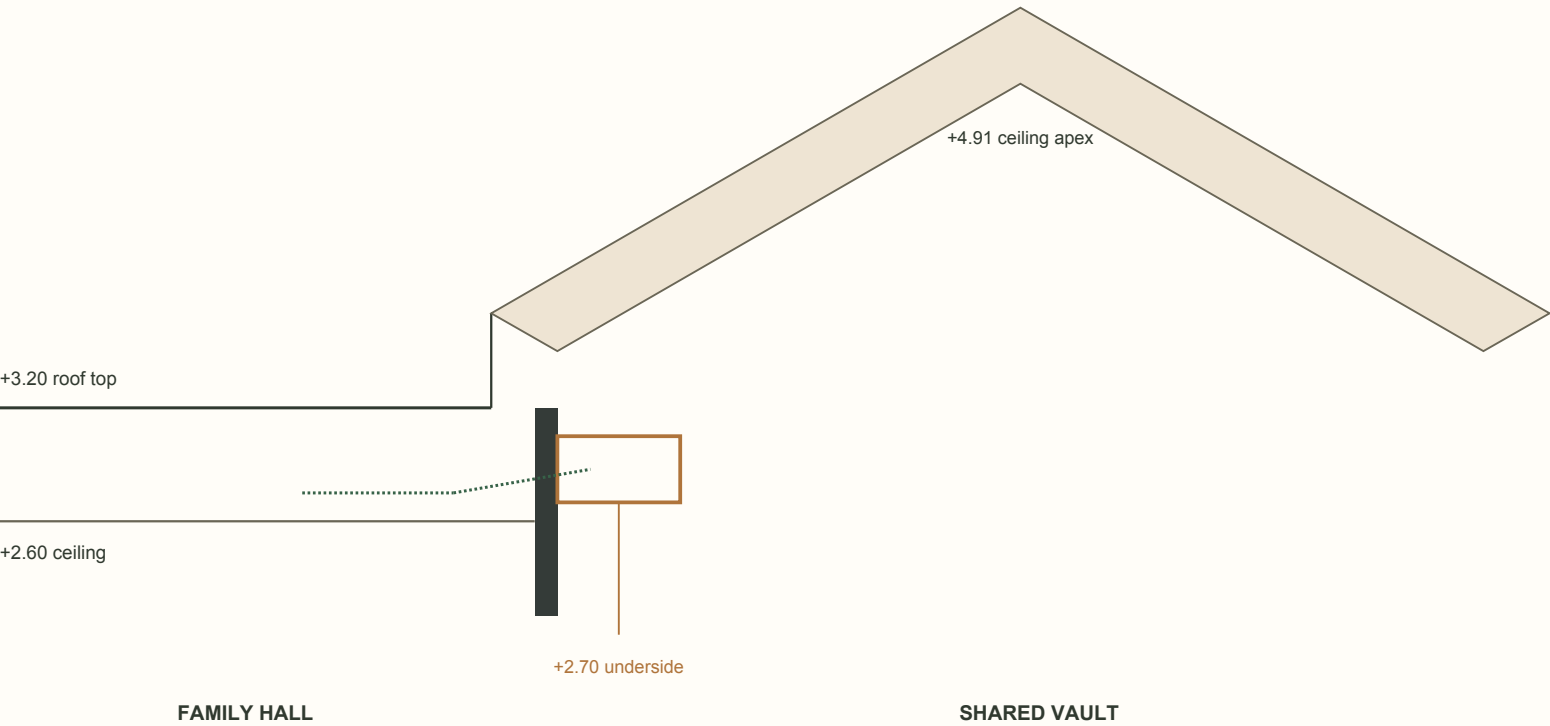
0.35 m base height + 0.85 m body = 1.20 m top.
Test 1.00 m vertical silencer reservations above the unit.
Then 0.30 m for bends; top at approximately 2.50 m.
Only 0.10 m remains below a 2.60 m ceiling.
These ancillary dimensions are study allowances, not selected parts.



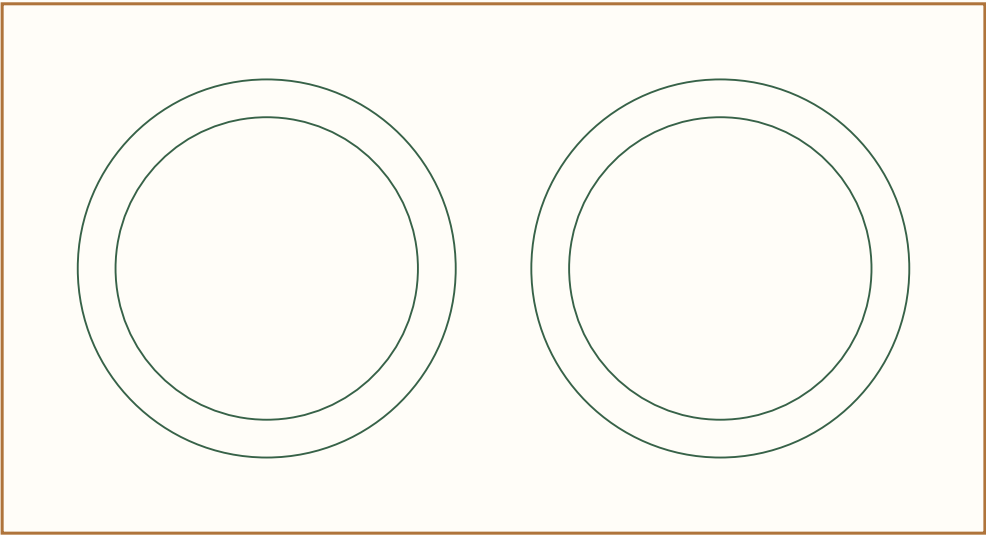
Front elevation; outdoor ducts omitted for clarity.

A central system leaves a substantial edge enclosure

Section through family hall and shared vault 1:40 at A3 | Duct packing detail 1:5 | Roof geometry retained as a concept



PAIR OF DUCTS / 1:5



PACKING ASSUMPTIONS

Two 200 mm nominal ducts, each with 25 mm insulation.
250 mm outer envelopes; 50 mm gap between them.
50 mm each side for lining / fixing / working allowance.
50 mm above and below; no spare space beyond this.
Ducts are NOT sized from pressure or airflow calculations.

WHAT STILL HAS TO FIT

The 13.05 m band runs between halls. Reserve larger local turn / junction zones; a straight packing section does not prove an elbow fits.
The lower wing has 0.60 m gross between roof top and ceiling. Structure, insulation and falls consume that depth; no clear void is assumed.
B uses local high-wall outlets near the wing boundaries instead. Their headers, grille depth, throw and acoustic performance still need design.
If local outlets cannot distribute fresh air well, B may still need short shared-room ducts; it is not a guaranteed invisible solution.

Develop the two-unit option, while keeping the trade-off explicit

No room, door or window changes adopted | All shelving gains are conditional on full relocation of the original plant

MEASURE / CONSEQUENCE	A / CENTRAL	B / TWO LOCAL
Main units / roof terminal pairs	1 / 1	2 / 2
ST linen run, 450 mm deep	2.05 m	1.25 m
Five-level shelf surface	4.61 m2	2.81 m2
Family-side unit bay	None	0.80 x 1.42 m
Through-running family trunks	13.05 m across shared edge	None in this proposal
Unit front maintenance space	1.00 m at rear	1.00 m at both
Key unresolved issue	Band, turns and roof/header fit	Local outlet throw and bedroom noise

SHARED-ROOM DUTIES UPDATED

Family unit: parents / child bedrooms plus the shared living end;
extract from parents' ensuite and family bathroom.
Service unit: guest, office, library, gym, garden and dining supply;
extract guest shower, WC, laundry and kitchen.
Both systems need balancing, including shared-room transfer effects.
Kitchen hood, plant and storage-space ventilation remain separate duties.

RESULTS AND NEXT PROOF

56 sampled daytime routes pass in each alternative.
Unit service rectangles clear fixed items; the open plant door clears routes.
Room geometry and all non-plant / linen furniture stay unchanged.
Roof pairs reserve 2.10 m rear / 2.00 m family spacing in plan.
Their height, windows, flues, PV and wind interaction are unresolved.
Prove outlet throw, noise, duty and complete duct fit before selection.

PRIMARY DIMENSION REFERENCES / CHECKED 13 SEPTEMBER 2026

[1] Q installer manual: access / drawings / terminals, pp. 5, 8-9, 16

[2] Current UK Q600 page: 725 x 570 x 850 mm body

Manufacturer dimensions ground the body/access examples. Bay sizes, ancillary zones, pipe packing and routing are this study's provisional design.